

Cybersecurity Awareness Training at a Regional 911 Dispatch Center

Findings and Recommendations

Prepared for: A regional 911 dispatch center (PSAP)
Prepared by: Michael Keuhlen June 2026

This summarizes the results of the cybersecurity awareness training delivered to dispatch staff on May 19–20, 2026. The training focused on the social engineering threats most relevant to a 911 environment: spear phishing, voice phishing (vishing), and vendor impersonation. Staff completed the same anonymous assessment before and after the session so the two could be compared directly.

The short version: the team started from a strong baseline, the training closed the remaining behavioral gaps on nearly every item, and it surfaced one specific judgment skill that is worth reinforcing.

What the assessment showed

Across the ten scenario questions, measured performance improved on every comparison point that moved, and held at full marks on the items the team already had down.

Measure	Before (n=26)	After (n=13)
Staff who missed at least one scenario question	34.6%	15.4%
Aggregate error rate across all scenario questions	3.8%	1.5%
Verifying a link before clicking (hover to preview the URL)	84.6%	100%
Independent callback verification	88.5%	84.6%

The overall behavioral error rate roughly halved, and the share of staff who missed any single scenario question dropped by more than half. Link verification, one of two soft spots at baseline, closed completely.

Confidence moved in step with skill rather than ahead of it. The two respondents who initially said they were not confident they could spot an attempt both moved up afterward, and no one reported low confidence or low familiarity post-training. Every post-training respondent also confirmed they know how to report a suspicious email, which had not been universal at baseline.

A note on the samples: 26 staff completed the pre-training assessment and 13 completed the post-training version. The lower follow-up count reflects timing. The post-training window overlapped with an active regional brush fire that raised call volume and operational workload. The post-training figures should therefore be read as directional rather than definitive, but the direction is consistent and clean across every measure.

The one gap worth attention

The single item that did not improve was independent callback verification. In that scenario, a caller claiming to be from a neighboring city's IT department asks the dispatcher to call them back at a number the caller provides. The correct response is to disregard the supplied number and independently look up and call the known IT number to confirm the request. Roughly the same share answered this correctly before and after the training (88.5% versus 84.6%).

This matters more than its size suggests, because “call me back at this number” is the exact mechanism behind most vendor impersonation and vishing attacks. A dispatcher who is helpful, fast, and used to coordinating across agencies, the same instincts that make someone good at the job, is the one most likely to accept a caller-provided number under pressure. This is the narrow skill where a short, targeted reinforcement would have the most value.

What this means for a capable team

The staff handled the more obvious threats well at both points: spoofed sender addresses, urgent account-lockout emails, requests to run unknown files, and basic reporting were answered correctly by nearly everyone before and after. The value of awareness training for this group is not in re-teaching fundamentals. It is in the small number of high-consequence judgment calls, independent verification and link handling, where a single wrong decision under pressure can matter.

One related data point: before this engagement, roughly a quarter of responding staff reported either no cybersecurity training in the past two years or were unsure. For participants, this engagement closed that gap, which is worth noting from a readiness and consistency standpoint.

Recommendations

These are offered as observations from the data rather than prescriptions, and they are intentionally low-lift.

1. **Make awareness training recurring, not one-time.** The training-recency gap at baseline, and the fact that felt readiness initially ran ahead of measured readiness, both argue for periodic reinforcement. A brief annual or semi-annual refresher would likely hold these gains.
2. **Add a short, specific focus on independent callback verification.** This is the one skill the data shows did not move. A two-minute reinforcement, a desk reference card, or a simple standing rule (never call a number a caller gives you to verify that caller; look it up yourself) would target it directly.
3. **Keep the reporting reflex visible.** Post-training, every respondent knew how to report a suspicious message. A periodic reminder of the exact reporting steps would keep that universal as staff turns over.
4. **Consider folding a short version into onboarding.** New hires arrive with the same helpful, high-trust instincts that social engineering targets. A brief module at onboarding would extend this baseline to every new dispatcher.

Thank you to the dispatch team for making this possible, and to everyone who gave their time and candor in participating. I am glad to walk through any of these findings in more detail.